TARUN SAHNAN

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Full Stack Engineer (3 YOE) - own the STAGE web platform end-to-end plus the systems around it: Next.js frontend, Node.js/NestJS backends, a self-hosted FFmpeg video transcoding pipeline, ClickHouse analytics, and AI agent systems. Ship outcomes, not just features.

SKILLS

Languages	JavaScript, TypeScript, SQL, HTML, CSS, Kotlin
Frontend	React.js, Next.js (App Router, SSR / SSG / ISR), Vue.js, Tailwind CSS, Material UI, PrimeVue, Storybook, Three.js
Backend & Data	Node.js, NestJS, Express.js, MongoDB, ClickHouse, REST, GraphQL
Video & Streaming	FFmpeg, HLS adaptive streaming (H.264 / H.265), hls.js, Mux, AWS MediaConvert, self-hosted transcoding
Infra & DevOps	AWS (S3, CloudFront, MediaConvert, Amplify), self-hosted bare-metal infra, GitHub Actions, CDN caching
Marketing & Attribution	AppsFlyer, Meta CAPI & Marketing API, GA4, Snapchat, Amplitude, DataDog
Performance & SEO	Web Vitals (LCP / CLS / INP), Lighthouse, bundle analysis; technical SEO, JSON-LD, programmatic SEO, Google Search Console
Testing & TV	Cypress (cypress-split), Jest, Vitest; Tizen OS (Samsung), webOS (LG), remote-key navigation, OTA
AI	Claude, Codex, Paperclip, AI-augmented development, multi-agent pipelines

EXPERIENCE

STAGE - *Software Engineer* **Jul 2024 - Present**

- **Solely developed the entire frontend of the STAGE website in Next.js (App Router)** with **SSR / SSG / ISR** - owning architecture, routing, data fetching, accessibility, E2E tests (Cypress, parallel-sharded CI under 8 minutes), and rollout from scratch.
- Currently building a **one-user-one-agent system** that gives every user a dedicated AI agent, owning its architecture and implementation end-to-end.
- Built an **in-house alternative to CleverTap** using **NestJS, PostgreSQL, ClickHouse, Kafka, and Temporal**, reducing monthly tooling costs by **INR 4 lakh**.
- Built a **self-hosted FFmpeg video transcoding pipeline on bare-metal servers** replacing AWS MediaConvert - **cut transcoding costs by ~70%**; designed no-downtime re-transcodes, gradual per-platform rollout, and a live ClickHouse health dashboard.
- **Owned event attribution end-to-end for the performance marketing team** across **AppsFlyer, Meta CAPI, GA4, and Snapchat**, making attribution the single source of truth for campaign spend decisions.
- Built a **self-hosted ads launcher for Meta and YouTube**, reducing ad launch time from **~30 min to ~5 min** with launch recovery and subscriber-performance reporting.
- Improved **Lighthouse performance score from ~58 to ~90** and reduced **LCP from 7s to 3s** via caching, image optimization, code-splitting, lazy loading, critical-path CSS, and bundle reduction.
- Owned the full **Web funnel**; improved **web-to-app install conversion from 27% to 42%** and drove a **2x improvement in subscription rate** end-to-end.
- Enhanced **SEO end-to-end**; built automated **SEO/GEO agents** with self-improving research that replaced the third-party SEO agency and drove 30+ PRs.
- Migrated **Next.js hosting from AWS Amplify to self-hosted infrastructure**, reducing infra costs by **~70%**.
- Shipped **Samsung TV and LG TV apps** and implemented **OTA updates**, reducing release time from **~2 months to ~2 minutes**.
- **Leading a team of 3 members** - driving technical direction, code reviews, unblocking, and Web production incident response.

Liberin Technologies - *Software Engineer* **Jun 2023 - Jun 2024**

- Built a high-performance **Vue 3 dashboard** from scratch using Vite, TypeScript, Tailwind CSS, and Vitest - owned architecture, component design, and rollout end-to-end.
- Created an **NPM package for multi-flow authentication**, standardizing login across internal projects and removing duplicated auth code.
- Increased **test coverage from 35% to 83%** in a legacy codebase; enhanced a custom **UI library** with dynamic components and Storybook-driven testing.
- Upgraded a Vue 2 dashboard for cross-platform use; designed and implemented **Node.js REST API services** and a Java-based web server.

Krishify - *Software Engineer Intern* **Jul 2022 - May 2023**

- Developed **"Bhumi"**, India's first Agri AI Chatbot - built in React within a 48-hour hackathon and shipped to production.
- Built a responsive **React-based E-Commerce platform** used by a multi-million-user farmer base; owned the frontend end-to-end.
- Optimized the **Android app in Kotlin** for performance and UX, improving load times and responsiveness.

PERSONAL PROJECTS

- **SWORD** *(in progress, open source)* - Native desktop GUI for Claude Code with a Claude.ai-inspired UI, markdown + Shiki highlighting, multi-chat sidebar, and a **zero-config auth flow**.
- **Citovo** *(in progress)* - Automated end-to-end **SEO & GEO platform**: SEO audits, programmatic SEO, AI-generated blogs, and agent-ready fix prompts.
- **Inspex.in** *(live, inspex.in)* - Company intelligence platform for Indian companies with AI chat assistance for audits and discovery.
- **Exam Sankalp** *(live, examsankalp.in)* - Practice platform for Indian government exams with **AI-generated questions**, previous-year banks, mock tests, and performance analytics.

EDUCATION

Lovely Professional University - *B.Tech, Computer Science* **2019 - 2023 · Phagwara, India · CGPA 8.54 / 10**